



CANDIDATE  
NAME

|  |
|--|
|  |
|--|

CENTRE  
NUMBER

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

CANDIDATE  
NUMBER

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

## 0620/61

October/November 2023

**1 hour**

You must answer on the question paper.

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [ ].
- Notes for use in qualitative analysis are provided in the question paper.

This document has **12** pages. Any blank pages are indicated.



- 1 A student uses chromatography to analyse samples of three different dyes. The apparatus the student uses is shown in Fig. 1.1.

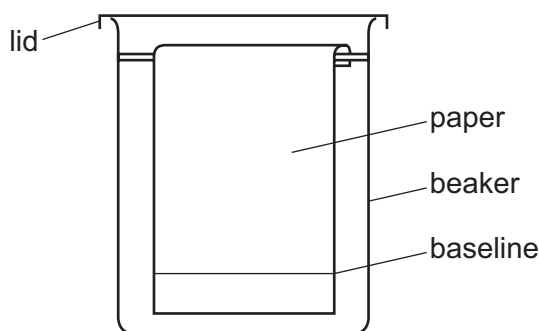


Fig. 1.1

- (a) A spot of each dye is placed on the paper and some ethanol is poured into the beaker.

Draw on Fig. 1.1:

- **three** spots (•) to show where the three dyes are placed on the paper at the start of the experiment
- a line to show the level of ethanol in the beaker at the start of the experiment.

[2]

- (b) During the experiment the ethanol moves up the paper.

State when the student should remove the chromatography paper from the ethanol in the beaker.

..... [1]

- (c) Fig. 1.2 shows the result of the chromatography experiment.

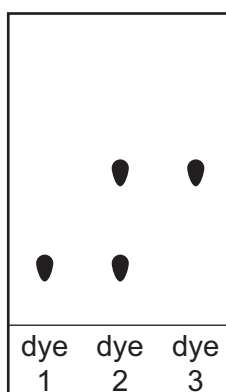


Fig. 1.2

State what conclusions can be made from this result.

.....  
 .....  
 ..... [2]

[Total: 5]

- 2 Copper(II) carbonate reacts with dilute acids to make carbon dioxide gas. Malachite is a mineral that contains copper(II) carbonate. A student investigates the rate of reaction between powdered malachite and dilute ethanoic acid at different temperatures. The student does six experiments.

Experiment 1

- Use a measuring cylinder to pour  $40\text{ cm}^3$  of dilute ethanoic acid into a conical flask.
- Warm the dilute ethanoic acid by about  $5^\circ\text{C}$ .
- Measure the temperature of the acid using a thermometer.
- Set the apparatus up as shown in Fig. 2.1.

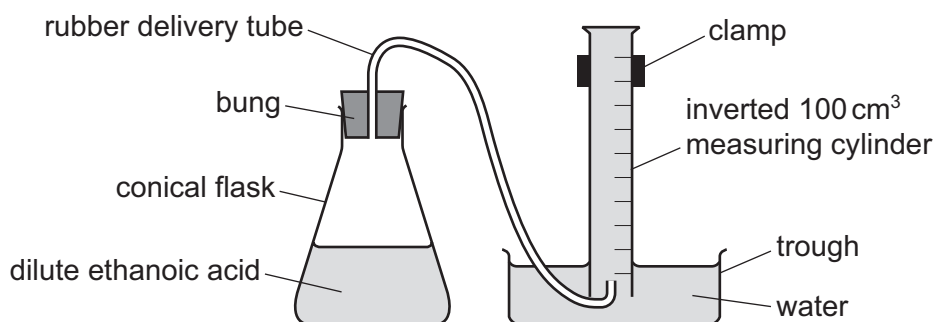


Fig. 2.1

- Remove the bung from the conical flask and add  $1.0\text{ g}$  of powdered malachite to the conical flask, replace the bung and start a stop-clock.
- Record the time taken for  $100\text{ cm}^3$  of gas to be collected in the measuring cylinder.
- Empty the conical flask and rinse it with distilled water.

Experiment 2

- Repeat Experiment 1 but warm the dilute ethanoic acid by about  $10^\circ\text{C}$ .

Experiment 3

- Repeat Experiment 1 but warm the dilute ethanoic acid by about  $15^\circ\text{C}$ .

Experiment 4

- Repeat Experiment 1 but warm the dilute ethanoic acid by about  $25^\circ\text{C}$ .

Experiment 5

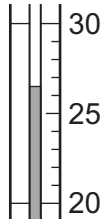
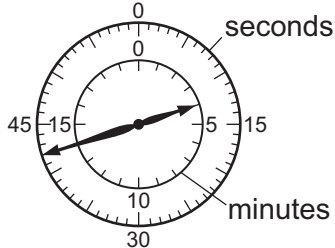
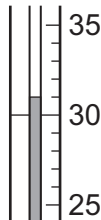
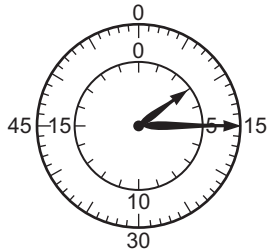
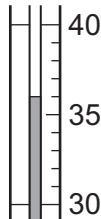
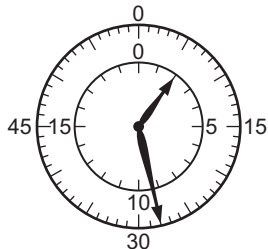
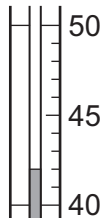
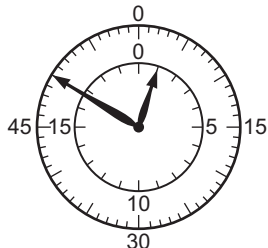
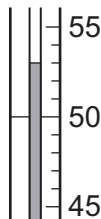
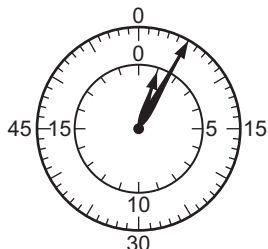
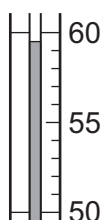
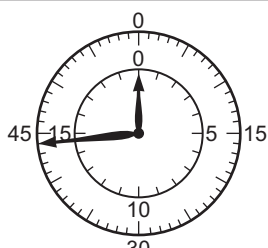
- Repeat Experiment 1 but warm the dilute ethanoic acid by about  $35^\circ\text{C}$ .

Experiment 6

- Repeat Experiment 1 but warm the dilute ethanoic acid by about  $40^\circ\text{C}$ .

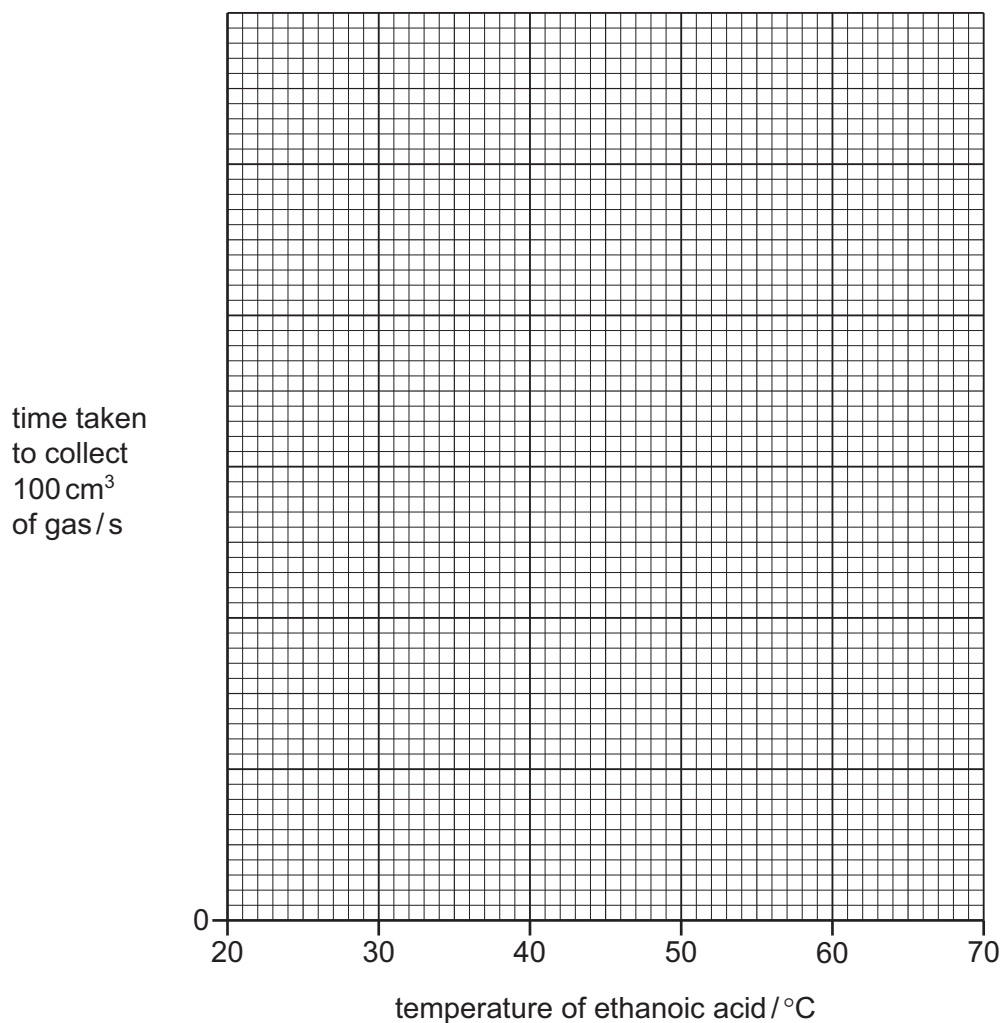
(a) Use the thermometer and stop-clock diagrams to complete Table 2.1.

**Table 2.1**

| experiment | thermometer diagram                                                                 | temperature of ethanoic acid / °C | stop-clock diagram                                                                   | time taken to collect 100 cm <sup>3</sup> of gas / s |
|------------|-------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------|
| 1          |    |                                   |    |                                                      |
| 2          |    |                                   |    |                                                      |
| 3          |   |                                   |   |                                                      |
| 4          |  |                                   |  |                                                      |
| 5          |  |                                   |  |                                                      |
| 6          |  |                                   |  |                                                      |

- (b) Complete a suitable scale on the y-axis and plot the results from Experiments 1 to 6 on Fig. 2.2.

Draw a line of best fit through your points.



**Fig. 2.2**

[4]

- (c) The average rate of reaction in an experiment can be calculated using the equation shown.

$$\text{average rate of reaction} = \frac{\text{volume of gas collected}}{\text{time taken to collect the gas}}$$

- (i) Use this equation to calculate the average rate of reaction in Experiment 6.  
Give the units for the rate you have calculated.

average rate = ..... units .....

[2]

- (ii) Deduce in which experiment, 1, 2, 3, 4, 5 or 6, the rate of reaction is greatest.

..... [1]

- (d) Extend the line on your graph in Fig. 2.2.  
Deduce the time taken to collect  $100\text{ cm}^3$  of gas when the temperature of the ethanoic acid is  $65^\circ\text{C}$ .

Show clearly on Fig. 2.2 how you worked out your answer.

..... S  
[3]

- (e) The  $40\text{ cm}^3$  of ethanoic acid used in each experiment is measured using a measuring cylinder. Measuring cylinders are available in the following sizes.

**$10\text{ cm}^3$        $25\text{ cm}^3$        $50\text{ cm}^3$        $100\text{ cm}^3$        $500\text{ cm}^3$**

Draw a circle around the size of measuring cylinder which would be most suitable to measure  $40\text{ cm}^3$  of ethanoic acid. [1]

- (f) Most of the gas collected in the measuring cylinder is air rather than carbon dioxide.

- (i) Explain why air is collected in the measuring cylinder.

.....  
..... [1]

- (ii) Explain why this does **not** affect the accuracy of the results.

.....  
..... [1]

- (g) During each experiment the temperature of the acid decreases slowly.

Give a reason why the temperature of the acid decreases and suggest a change to the apparatus used that would minimise the decrease in temperature.

reason temperature decreases .....  
.....  
change to apparatus .....  
..... [2]

[Total: 20]

- 3 A student tests two substances: solid **M** and solid **N**.

**Tests on solid M**

Solid **M** is hydrated chromium(III) nitrate.

Complete the expected observations.

- (a) The student places half of solid **M** in a boiling tube and heats it strongly.

- (i) The student holds a piece of anhydrous cobalt(II) chloride paper at the mouth of the boiling tube.

observations .....

..... [1]

- (ii) The student inserts a glowing splint into the mouth of the boiling tube. The splint bursts into flames.

Identify the gas given off by heating solid **M** which causes this result.

..... [1]

- (b) The student dissolves the remaining solid **M** in water to form solution **M**.

- (i) The student adds aqueous sodium hydroxide dropwise and then in excess to solution **M**.

observations adding dropwise .....

observations in excess .....

[2]

- (ii) The student adds a piece of aluminium foil to the product from (b)(i). The mixture is then warmed. The student tests for any gas produced.

observations .....

.....

..... [1]

**Tests on solid N**

Table 3.1 shows the tests and the student's observations for solid **N**.

**Table 3.1**

| tests                                                                                                                                                                                                                        | observations            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>test 1</b><br><br>Do a flame test on solid <b>N</b> .                                                                                                                                                                     | light green flame       |
| <b>test 2</b><br><br>Dissolve the remaining solid <b>N</b> in water to form solution <b>N</b> .<br>Divide solution <b>N</b> into four portions.<br><br>To the first portion of solution <b>N</b> , add dilute sulfuric acid. | white precipitate       |
| <b>test 3</b><br><br>To the second portion of solution <b>N</b> , add about 1 cm <sup>3</sup> of dilute nitric acid followed by a few drops of aqueous silver nitrate.                                                       | pale yellow precipitate |
| <b>test 4</b><br><br>To the third portion of solution <b>N</b> , add about 1 cm <sup>3</sup> of dilute nitric acid followed by a few drops of aqueous barium nitrate.                                                        | no visible change       |
| <b>test 5</b><br><br>To the fourth portion of solution <b>N</b> , add about 1 cm <sup>3</sup> of aqueous chlorine.                                                                                                           | brown solution          |

(c) Describe how to carry out a flame test.

.....

.....

..... [2]

(d) Identify solid **N**.

.....

..... [2]

[Total: 9]

- 4 *Greaseaway* and *Kitchenclean* are two solutions used as household cleaners that contain aqueous ammonia.

Plan an investigation to find which of the two household cleaners contains aqueous ammonia with the highest concentration. Assume that aqueous ammonia is the only alkali in the cleaners.

Include in your plan:

- the method you will use
- how your results will be used to determine which household cleaner contains aqueous ammonia with the highest concentration.

You are provided with common laboratory apparatus and chemicals.

[6]



## Notes for use in qualitative analysis

### Tests for anions

| anion                                        | test                                                              | test result                                                                             |
|----------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| carbonate, $\text{CO}_3^{2-}$                | add dilute acid, then test for carbon dioxide gas                 | effervescence, carbon dioxide produced                                                  |
| chloride, $\text{Cl}^-$<br>[in solution]     | acidify with dilute nitric acid, then add aqueous silver nitrate  | white ppt.                                                                              |
| bromide, $\text{Br}^-$<br>[in solution]      | acidify with dilute nitric acid, then add aqueous silver nitrate  | cream ppt.                                                                              |
| iodide, $\text{I}^-$<br>[in solution]        | acidify with dilute nitric acid, then add aqueous silver nitrate  | yellow ppt.                                                                             |
| nitrate, $\text{NO}_3^-$<br>[in solution]    | add aqueous sodium hydroxide, then aluminium foil; warm carefully | ammonia produced                                                                        |
| sulfate, $\text{SO}_4^{2-}$<br>[in solution] | acidify with dilute nitric acid, then add aqueous barium nitrate  | white ppt.                                                                              |
| sulfite, $\text{SO}_3^{2-}$                  | add a small volume of acidified aqueous potassium manganate(VII)  | the acidified aqueous potassium manganate(VII) changes colour from purple to colourless |

### Tests for aqueous cations

| cation                          | effect of aqueous sodium hydroxide                                         | effect of aqueous ammonia                                                  |
|---------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| aluminium, $\text{Al}^{3+}$     | white ppt., soluble in excess, giving a colourless solution                | white ppt., insoluble in excess                                            |
| ammonium, $\text{NH}_4^+$       | ammonia produced on warming                                                | —                                                                          |
| calcium, $\text{Ca}^{2+}$       | white ppt., insoluble in excess                                            | no ppt. or very slight white ppt.                                          |
| chromium(III), $\text{Cr}^{3+}$ | green ppt., soluble in excess                                              | green ppt., insoluble in excess                                            |
| copper(II), $\text{Cu}^{2+}$    | light blue ppt., insoluble in excess                                       | light blue ppt., soluble in excess, giving a dark blue solution            |
| iron(II), $\text{Fe}^{2+}$      | green ppt., insoluble in excess, ppt. turns brown near surface on standing | green ppt., insoluble in excess, ppt. turns brown near surface on standing |
| iron(III), $\text{Fe}^{3+}$     | red-brown ppt., insoluble in excess                                        | red-brown ppt., insoluble in excess                                        |
| zinc, $\text{Zn}^{2+}$          | white ppt., soluble in excess, giving a colourless solution                | white ppt., soluble in excess, giving a colourless solution                |

**Tests for gases**

| gas                           | test and test result                                                       |
|-------------------------------|----------------------------------------------------------------------------|
| ammonia, $\text{NH}_3$        | turns damp red litmus paper blue                                           |
| carbon dioxide, $\text{CO}_2$ | turns limewater milky                                                      |
| chlorine, $\text{Cl}_2$       | bleaches damp litmus paper                                                 |
| hydrogen, $\text{H}_2$        | 'pops' with a lighted splint                                               |
| oxygen, $\text{O}_2$          | relights a glowing splint                                                  |
| sulfur dioxide, $\text{SO}_2$ | turns acidified aqueous potassium manganate(VII) from purple to colourless |

**Flame tests for metal ions**

| metal ion                    | flame colour |
|------------------------------|--------------|
| lithium, $\text{Li}^+$       | red          |
| sodium, $\text{Na}^+$        | yellow       |
| potassium, $\text{K}^+$      | lilac        |
| calcium, $\text{Ca}^{2+}$    | orange-red   |
| barium, $\text{Ba}^{2+}$     | light green  |
| copper(II), $\text{Cu}^{2+}$ | blue-green   |

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at [www.cambridgeinternational.org](http://www.cambridgeinternational.org) after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.